

Power Sums, Newton's Identities, and the Pandrosion Field

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Abstract

The Pandrosion field $F_P(z) = P'(z)/P(z) = \sum 1/(z - \alpha_j)$ is the generating function of all power sums: $F_P(z) = n/z + p_1/z^2 + p_2/z^3 + \dots$ where $p_k = \sum \alpha_j^k$. Newton's identities $p_k - e_1 p_{k-1} + \dots + (-1)^{k-1} k e_k = 0$ follow in one line from $F_P \cdot P = P'$ (coefficient matching). This closes the Pandrosion cycle: $F_P \rightarrow p_k \rightarrow e_j \rightarrow \text{Turán SOS (Paper 56)} \rightarrow \text{Newton inequality (Paper 58)} \rightarrow \text{log-concavity (Paper 66)}$.

1 The Pandrosion field as generating function

Theorem 1.1 (Laurent expansion of F_P). *For $P(z) = \prod (z - \alpha_j)$, with $p_k = \sum_{j=1}^n \alpha_j^k$:*

$$\boxed{F_P(z) = \frac{P'(z)}{P(z)} = \frac{n}{z} + \frac{p_1}{z^2} + \frac{p_2}{z^3} + \dots = \sum_{k=0}^{\infty} \frac{p_k}{z^{k+1}},} \quad (1)$$

where $p_0 = n$. Equivalently, $z F_P(z) = n + p_1/z + p_2/z^2 + \dots$

Proof. $F_P(z) = \sum_j 1/(z - \alpha_j) = \sum_j (1/z)(1/(1 - \alpha_j/z)) = \sum_j (1/z) \sum_{k \geq 0} (\alpha_j/z)^k = \sum_{k \geq 0} p_k/z^{k+1}$. $\square$

2 Newton's identities from $F_P \cdot P = P'$

Theorem 2.1 (One-line proof of Newton's identities). *All Newton identities, for $1 \leq k \leq n$:*

$$p_k - e_1 p_{k-1} + e_2 p_{k-2} - \dots + (-1)^{k-1} k e_k = 0, \quad (2)$$

follow simultaneously from the single equation $F_P(z) \cdot P(z) = P'(z)$.

Proof. Write $P(z) = z^n - e_1 z^{n-1} + \dots + (-1)^n e_n$ and $F_P(z) = \sum_{k \geq 0} p_k z^{-(k+1)}$. Then:

$$F_P \cdot P = \left(\sum_{k \geq 0} p_k z^{-(k+1)} \right) (z^n - e_1 z^{n-1} + \dots).$$

The coefficient of z^{n-1-m} on the left is $\sum_{j=0}^m (-1)^j e_j p_{m-j}$ (with $e_0 = 1$).

On the right, $P' = n z^{n-1} - (n-1) e_1 z^{n-2} + \dots$, so the coefficient of z^{n-1-m} is $(n-m)(-1)^m e_m$.

Matching gives exactly Newton's identity for index $k = m$. Since this holds for all m , every Newton identity is extracted from the single equation. $\square$

3 Waring's formula

By solving the Newton recursion from Theorem 2.1:

$$p_1 = e_1, \quad (3)$$

$$p_2 = e_1^2 - 2e_2, \quad (4)$$

$$p_3 = e_1^3 - 3e_1e_2 + 3e_3, \quad (5)$$

$$p_4 = e_1^4 - 4e_1^2e_2 + 2e_2^2 + 4e_1e_3 - 4e_4. \quad (6)$$

These are the *Waring formulas*—each p_k as a polynomial in the elementary symmetric functions—derived directly from the Pandrosion field expansion.

4 The complete Pandrosion cycle

Theorem 4.1 (Closed cycle). *The following chain is driven entirely by $F_P = P'/P$:*

$$\underbrace{F_P}_{\text{field}} \xrightarrow{\text{Laurent}} \underbrace{p_k}_{\text{power sums}} \xrightarrow{\text{Newton}} \underbrace{e_j}_{\text{elem. sym.}} \xrightarrow{\text{Paper 56}} \underbrace{\sum Q_j^2}_{\text{Turán SOS}} \xrightarrow{\text{Paper 58}} \underbrace{e_k^2 \geq \dots}_{\text{log-concavity}} \quad (7)$$

Object	Paper	Derives from F_P
Power sums p_k	This paper (70)	Laurent coefficients
Newton identities	This paper (70)	$F_P \cdot P = P'$
Turán SOS	Paper 56	$F_P^2 \cdot P^2 - P \cdot (F_P P)' = \sum Q^2$
Log-concavity	Paper 58	Cauchy-Schwarz on SOS
Hermite-Biehler	Paper 65	$\text{Im } F_P > 0$
Stability	Paper 67	$\text{Re } F_P > 0$ on boundary
Bézout	Paper 69	$F_P - F_Q$ resonances

References

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- [4] I. Besevic, *Log-concavity from the Pandrosion sum of squares*, Pandrosion Dynamics **58** (2026).